

QUESTION BANK WITH ANSWERS

Dr. Zinab — Full Answer Key

Sections: Renal · Burns · Endocrine · Neuro/Infectious

Total MCQs: 32 | Essays: 10

KEY: Correct answers are shown in **green** with the answer letter highlighted in a green box after each question.

SECTION A — MULTIPLE CHOICE QUESTIONS (32 Questions)

■ Renal System

1. Acute renal failure is best defined as:

✓ B

- A) Gradual loss of kidney function over years
- C) Chronic decrease in urine output

- B) Sudden loss of kidney function over hours to days ✓**
- D) Permanent destruction of nephrons

2. Chronic renal failure (ESRD) is best defined as:

✓ D

- A) Sudden loss of kidney function over hours to days
- C) Temporary decline in glomerular filtration rate (GFR)

- B) Acute tubular necrosis due to nephrotoxins
- D) Progressive, irreversible deterioration in renal function ✓**

3. Pre-renal causes of ARF are primarily due to:

✓ C

- A) Obstruction of urinary outflow
- C) Impaired blood flow leading to hypoperfusion ✓**

- B) Parenchymal damage to kidney tissue
- D) Genetic abnormalities of the nephron

4. Post-renal causes of ARF are usually related to:

✓ B

- A) Decreased cardiac output
- C) Sepsis-induced vasodilation

- B) Obstruction distal to the kidney ✓**
- D) Hypotension

5. Bed rest during the acute stage of ARF is indicated primarily to:

✓ B

- A) Prevent infection
- C) Improve pulmonary function

- B) Reduce exertion ✓**
- D) Promote skin integrity

6. Which is the most accurate method to monitor fluid status in a patient with ARF?

✓ C

A) Checking blood pressure once daily
C) **Accurate daily weights and intake/output records ✓**

B) Measuring urine color
D) Observing skin turgor only

7. Acute dialysis is indicated when the patient has:

✓ B

A) Mild dehydration
C) Stable electrolyte balance

B) **High serum potassium ✓**
D) Normal renal function

8. Chronic or maintenance dialysis is primarily indicated in:

✓ A

A) **End-stage renal disease ✓**
C) Temporary electrolyte imbalance

B) Acute renal failure
D) Mild renal impairment

9. The most commonly used method of dialysis is:

✓ B

A) Peritoneal dialysis
C) Oliguria management

B) **Hemodialysis ✓**
D) Ultrafiltration only

10. Long-term peritoneal dialysis can lead to which metabolic complication?

✓ B

A) Hypoglycemia
C) Hypocalcemia

B) **Hypertriglyceridemia ✓**
D) Hyponatremia

11. Which of the following is a classic urinary finding in glomerulonephritis?

✓ B

A) Polyuria
C) Glycosuria

B) **Hematuria ✓**
D) Pyuria

12. The most common cause of pyelonephritis is:

✓ B

A) Hematogenous spread of bacteria
C) Viral infection of the kidney

B) **Ascending infection from the bladder ✓**
D) Autoimmune reaction

13. Peak incidence of renal stones occurs at which age group?

✓ C

A) 10-20 years
C) **30-50 years ✓**

B) 20-30 years
D) 60-80 years

14. Renal stones affect approximately what percentage of the population?

✓ B

A) 1-5%
C) 20-25%

B) **10-15% ✓**
D) 30-40%

■ Burns

15. A factory worker sustains a burn after accidental contact with sulfuric acid. This is classified as:

✓ B

- A) Thermal burn
- B) Chemical burn ✓
- C) Electrical burn
- D) Radiation burn

16. Which of the following is NOT a typical cause of burn injury?

✓ C

- A) Heat (flames, hot liquids)
- B) Electricity
- C) Hypothermia ✓
- D) Chemicals

17. Which best describes a first-degree burn?

✓ B

- A) Damage extends through dermis and subcutaneous tissue
- B) Only the epidermis is affected, with redness and pain ✓
- C) Involves destruction of muscles and bones
- D) Causes blistering and damage to both epidermis and dermis

18. Which type of burn is typically painless due to nerve destruction?

✓ D

- A) First-degree burn
- B) Superficial partial-thickness burn
- C) Deep partial-thickness burn
- D) Full-thickness burn ✓

■ Endocrine System

19. Type 2 Diabetes Mellitus is primarily characterized by:

✓ B

- A) Absolute lack of insulin secretion
- B) Insulin resistance with relative insulin deficiency ✓
- C) Complete resolution after delivery
- D) Genetic absence of pancreatic tissue

20. Which of the following is a classic symptom of diabetes mellitus?

✓ C

- A) Weight gain
- B) Hypotension
- C) Polyuria ✓
- D) Bradycardia

21. Gestational Diabetes Mellitus is defined as:

✓ B

- A) Diabetes diagnosed before pregnancy
- B) Diabetes that develops during pregnancy ✓
- C) Type 1 diabetes occurring in adolescents
- D) Secondary diabetes due to pancreatic disease

22. Which type of diabetes may resolve spontaneously after delivery?

✓ C

- A) Type 1 diabetes mellitus
- B) Type 2 diabetes mellitus
- C) Gestational diabetes mellitus ✓
- D) Secondary diabetes mellitus

23. Which of the following is a common clinical manifestation of hyperthyroidism?

✓ B

- A) Bradycardia
C) Cold intolerance

- B) Nervousness ✓
D) Weight gain

24. Pulse rate in hyperthyroidism typically ranges between:

✓ C

- A) 60-80 beats per minute
C) 90-160 beats per minute ✓

- B) 70-90 beats per minute
D) 180-200 beats per minute

25. If hypothyroidism is due to iodine deficiency, the client's diet should be:

✓ B

- A) Low in iodine
C) Free of fiber

- B) High in iodine ✓
D) High in soy products

26. How should thyroid medication be taken for optimal utilization?

✓ B

- A) With milk or calcium supplements
C) With high-fiber meals

- B) On an empty stomach ✓
D) Immediately after eating cruciferous vegetables

27. Cushing's syndrome results from:

✓ D

- A) Deficient adrenocortical activity
C) Low cortisol levels

- B) Reduced ACTH secretion
D) Excessive adrenocortical activity ✓

28. Patients with Cushing's syndrome are at increased risk for:

✓ A

- A) Infections ✓
C) Cold intolerance

- B) Hypotension
D) Hypothyroidism

■ Neurological & Infectious Diseases

29. Multiple Sclerosis (MS) is best described as:

✓ D

- A) A degenerative disorder of peripheral nerves caused by acetylcholine deficiency
C) A genetic condition leading to dopamine deficiency in the basal ganglia

- B) A progressive loss of motor neurons in the spinal cord and brainstem
D) A chronic, progressive autoimmune disorder attacking the myelin sheath in the CNS ✓

30. Systemic Lupus Erythematosus (SLE) is best described as:

✓ A

- A) A chronic inflammatory disorder of connective tissue affecting multiple organ systems ✓
C) A degenerative joint disease

- B) A bacterial infection of the skin
D) A localized autoimmune condition

31. Which of the following is often one of the first signs of HIV infection?

✓ C

- A) Skin rash
C) Swollen lymph nodes ✓

- B) Oral ulcers
D) Night sweats

32. Which of the following is a major route of HIV transmission?

✓ B

A) Sharing food and utensils

B) Unprotected sexual intercourse ✓

C) Mosquito bites

D) Handshakes

SECTION B — ESSAY QUESTIONS WITH FULL ANSWERS

1. Nursing Interventions for a Patient with Burns

[Key Nursing Actions]

- ◆ Respiratory Care: Ensure patent airway; provide humidified oxygen and maintain oxygenation through proper positioning and secretion removal.
- ◆ Fluid & Electrolyte Balance: Monitor serum electrolytes, daily weights, and hourly intake/output (I&O;).
- ◆ Infection Control: Wash hands before all patient contact; use protective gowns, gloves, and masks; apply prescribed topical antibacterials to the wound.
- ◆ Nutrition: Provide a high-calorie, high-protein diet; give supplemental vitamins and minerals as prescribed.
- ◆ Pain Management: Carefully assess pain; offer analgesics and relaxation techniques.
- ◆ Body Temperature: Maintain a warm environment and monitor rectal temperature regularly.
- ◆ GI Function: Auscultate for bowel sounds every four hours; test gastric content for bleeding.

2. Complications of Acute Renal Failure (ARF)

- ◆ Hyperkalemia: Declined GFR prevents normal potassium excretion, raising serum potassium to dangerous levels.
- ◆ Metabolic Acidosis: Inability to eliminate the daily metabolic load of acid-type substances leads to acid buildup.
- ◆ Calcium & Phosphorus Abnormalities: Serum phosphate rises while calcium levels drop, risking bone disease.
- ◆ Anemia: Results from reduced erythropoietin production and potential gastrointestinal blood loss.

3. Types of Diabetes Mellitus

- ◆ Type 1: Characterized by loss of insulin-producing beta cells, leading to absolute insulin deficiency.
- ◆ Type 2: Results from insulin resistance where cells fail to use insulin properly.
- ◆ Gestational DM: High blood sugar first recognized during pregnancy; may resolve after delivery.
- ◆ Secondary Diabetes: Caused by pancreatic diseases, hormonal abnormalities, or drugs such as corticosteroids.

4. Complications of Graves' Disease

- ◆ Exophthalmos: Protruding or bulging eyes due to orbital tissue swelling.
- ◆ Heart Disease: Cardiac complications including atrial fibrillation and heart failure from excessive thyroid hormones.
- ◆ Thyroid Storm: A severe, life-threatening condition (thyrotoxic crisis) requiring immediate intervention.

5. Categories of Acute Renal Failure

- ◆ Pre-renal (55-60%): Caused by hypoperfusion of the kidney — often due to heart failure, sepsis, or hypovolemia.
- ◆ Intra-renal (5%): Results from actual damage to kidney tissue caused by burns, infections, or nephrotoxic agents.
- ◆ Post-renal (35-40%): Caused by obstruction to urine flow such as kidney stones or tumors.

6. Complications of Chronic Renal Failure (ESRD)

- ◆ Cardiovascular: Hypertension, heart failure, and pulmonary edema.
- ◆ Hematologic: Severe anemia due to inadequate erythropoietin production.
- ◆ Metabolic: Hyperkalemia and metabolic acidosis.
- ◆ Musculoskeletal: Bone disease (renal osteodystrophy) and metastatic calcifications.
- ◆ Other: Pericarditis and severe uremic pruritus (itching).

7. Characteristics of a Moderate Uncomplicated Burn

- ◆ Second-degree burns: Involving 15%-25% TBSA in adults, or 10%-20% TBSA in children.
- ◆ Third-degree burns: Less than 10% TBSA not involving the face, hands, feet, or genitalia.
- ◆ Exclusions: Does not include electrical injuries, inhalation injuries, or concurrent major trauma.

8. Risk Factors of Multiple Sclerosis (MS)

- ◆ Age & Gender: Most common between 20-40 years; 2-3 times more common in females.
- ◆ Genetics & Ethnicity: Family history of MS and Northern European descent increase risk.
- ◆ Environment: Vitamin D deficiency, living in temperate climates, and prior viral infections.
- ◆ Lifestyle: Smoking is a known modifiable risk factor.

9. Nurse's Role for a Patient with Pyelonephritis

[Monitoring]

- ◆ Check vital signs regularly; assess flank/back pain; monitor I&O; and urine characteristics (color and odor).

[Medication]

- ◆ Administer prescribed antibiotics, antipyretics for fever, and analgesics for pain management.

[Hydration]

- ◆ Encourage high intake of oral fluids; administer IV fluids as needed to flush the urinary tract.

[Patient Education]

- ◆ Teach the importance of completing the full antibiotic course, proper hygiene (wiping front to back), and urinating after sexual intercourse.

10. Complications of Hemodialysis

◆	Cardiovascular: Atherosclerosis, heart failure, angina, and hypotension during dialysis sessions.
◆	Hematologic: Anemia and persistent fatigue.
◆	Musculoskeletal: Bone pain and fractures due to renal osteodystrophy.
◆	Gastrointestinal: Gastric ulcers and other GI problems including nausea and vomiting.
◆	Other: Muscle cramping, sleep disturbances, and uremic pruritus (itching).

— End of Question Bank — Dr. Zinab —